CNuvS Ex. 12 - Jan Lindauer, Paul Feidieker

Task 3

(a) A peer to peer system is a fully decentralized system where there are no servers normally. Contrary to a client/server system in P2P the clients function as clients and servers at the same time. In P2P every client benefits from the resources of the other participating clients.
[1, p. 6]

(b) Gnutella is a file transfer protocol. A sender sends a query message over all existing edges (valid TCP connections). Every peer forwards the query message to all it's edges and returns a query hit over the reverse path if it got the desired data.
[1, p. 11, 12]

(c) The churn problem refers to the continuous, rapid, and unpredictable joining and leaving of nodes in the network.
This can cause data loss (if the leaving note holds data that other nodes may request) and inefficient routing since routes over the leaving node become inaccessible.
[2]

Task 4

(a) No, the absolute clock values are not comparable. The durations are.
[3, p. 8]

(b) P sets it's clock to 4960.

This results from: $T_{S_2} + \frac{1}{2} * \text{message transmission delay}$

Message transmission delay:
$\delta = \delta_{\text{req}} + \delta_{\text{resp}} = \left(T_{C_2} - T_{C_1}\right) - \left(T_{S_2} - T_{S_1}\right)$
$\delta = (4920 - 4200) - (4700 - 4500)$
$\delta = 720 - 200$
$\delta = 520$

Clock set = $T_{S_2} + \frac{1}{2} * \delta$
Clock set = $4700 + \frac{1}{2} * 520$
Clock set = $4700 + 260$
Clock set = 4960
[3, p. 9]

(c)

Task 5

(a)

(b)

Sources

[1] T. D. Prof. Dr. Marco Zimmerling, "CNuvS 07: Application Layer." 2026.

[2] R. R. Daniel Stutzbach, "Understanding Churn in Peer-to-Peer Networks." [Online]. Available: https://ix.cs.uoregon.edu/~reza/PUB/churn.pdf

[3] Zsolt Istvan, "Time in Distributed Systems, Primary-Backup Replication." 2026.